

SLIDRONIX

AI-Based Landslide Early Warning System for the Western Ghats

Complete Implementation & Methodology Report — Phase A to Phase B

Project status covered: Phase A — Data Preparation & Baseline Model; Phase B — Terrain Feature Extraction

Purpose: This document records the implementation completed so far, the datasets created, algorithms used, validation results, limitations, and reproducibility details before continuing to Phase C/D.

1. Project Overview

Slidronix is being developed as a multimodal AI-based landslide risk and early-warning system focused on the Western Ghats. The planned system combines historical landslide inventory, terrain information, satellite-derived environmental features, soil properties, rainfall history and temporal rainfall sequences. The final architecture is intended to combine spatial, temporal and tabular information and produce a landslide-risk probability that can be visualized on a GIS map.

Roadmap followed

Phase	Main work	Status
A	Data preparation, cleaning, EDA, Western Ghats study region, binary dataset, Random Forest	Completed
B	DEM-based elevation and slope extraction	Completed
C	NDVI, land cover, soil	In progress / next
D	Historical rainfall and temporal rainfall sequences	Next
E	Dataset expansion / balanced spatial dataset	Largely completed through background sampling
F	CNN + temporal Transformer + tabular encoder + fusion	Planned
G	Model comparison and evaluation	Planned
H	GIS risk map and early-warning/alert dashboard	Planned

2. Phase A — Data Preparation & Baseline Model

2.1 Historical Landslide Inventory

The starting inventory was a 582-page field-validated landslide report for India. The PDF was processed programmatically using pdfplumber to extract tabular records. The extracted dataset contains 34,207 records and 11 attributes.

Attribute	Description
S.No	Record identifier
Latitude / Longitude	Geographic coordinates
Slide_Name	Landslide/site name where available
State	State of the record
District	District
Subdivision Or Taluk	Administrative subdivision
Material Involved	Recorded material
Movement Type	Historical movement classification

Attribute	Description
Initiation_Year	Reported initiation year, where available
History_date	Historical date, where available

2.2 Extraction and Cleaning Pipeline

The raw table was converted into a structured CSV. Latitude and longitude were converted to numeric values and restricted to plausible Indian coordinates. Categorical text fields were stripped/standardized. Initiation_Year was converted to numeric form, with zero treated as missing. Duplicate rows were removed at the row level. Records missing essential target/location/material information were removed.

Primary data products

File / output	Purpose
data/processed/landslide_inventory_raw.csv	PDF-extracted inventory
data/processed/landslide_inventory_cleaned.csv	Cleaned historical inventory
src/extract_dataset.py	Extraction pipeline
src/preprocess_data.py	Cleaning/preprocessing
src/inspect_dataset.py	Dataset inspection

2.3 Movement-Type Distribution

The historical inventory is strongly imbalanced. The dominant class is Slide, with much smaller numbers of Flow, Fall, Subsidence and other movement types. This imbalance became important when interpreting the Random Forest baseline: overall accuracy alone is not representative of minority-class performance.

Movement type	Count
Slide	30,040
Fall	1,541
Flow	1,405
Subsidence	567
Composite	124
Topple	81
slide	75
Slump	72
Creep	52
Complex	33

2.4 Western Ghats Study Region

The study region was constructed around the Western Ghats states represented in the project inventory: Maharashtra, Goa, Karnataka, Kerala and Tamil Nadu. A spatial envelope was derived using administrative district boundaries and the historical inventory. The resulting study polygon covers approximately 177,853.97 km² and contains 7,884 of the 7,885 selected positive landslide records.

Important terminology: this boundary should be described in reports as “Western Ghats study boundary derived from administrative boundaries and the historical landslide inventory.” It should not be presented as an official Western Ghats boundary unless replaced later by an authoritative Western Ghats polygon.

State	Positive landslide records
Maharashtra	1,370

State	Positive landslide records
Goa	68
Karnataka	2,274
Kerala	3,282
Tamil Nadu	891
Total	7,885

2.5 Positive + Background Binary Dataset

For a true binary landslide-risk formulation, historical positive points alone are insufficient because they contain no explicit no-landslide/background class. A background sampling procedure was therefore created inside the study polygon. Exactly 7,885 background points were generated to balance the 7,885 positive records.

Background sampling safeguards

Check	Result
Positive samples	7,885
Background samples	7,885
Total binary samples	15,770
Background duplicates	0
Background-positive overlap	0
Minimum positive separation	1.0025 km
Mean nearest-positive distance	13.6210 km
Maximum nearest-positive distance	76.1542 km
Study area	177,853.97 km ²

The minimum 1 km separation was enforced geodesically using a Haversine BallTree. This reduces the chance that a sampled background point is immediately adjacent to a known positive point.

Important limitation

The background points are sampled non-event locations, not independently confirmed absence-of-landslide observations. They are therefore best described as background samples rather than guaranteed true negatives.

2.6 Historical Baseline Random Forest

The first machine-learning baseline was deliberately simple and was used as a reference model. It predicts historical **Movement Type**, not future landslide occurrence. Features were Latitude, Longitude, State and Material Involved. Numeric missing values were median-imputed; categorical missing values were most-frequent imputed; categorical features were one-hot encoded.

```
RandomForestClassifier(
    n_estimators=150,
    max_depth=20,
    min_samples_leaf=2,
    class_weight="balanced",
    random_state=42,
    n_jobs=-1
)
```

An 80/20 stratified train-test split produced 27,183 training samples and 6,796 test samples.

Metric	Result
Accuracy	69.60%

Metric	Result
Macro F1	0.33
Weighted F1	0.76
Interpretation	Accuracy/weighted F1 are strongly affected by the dominant Slide class; macro F1 exposes weak minority-

The baseline is intentionally not treated as the final early-warning model. The original inventory does not contain sufficient negative samples or dynamic environmental triggers for a legitimate binary early-warning formulation.

3. Phase B — Terrain Feature Extraction

3.1 Why Terrain Features Were Added

The project then moved from historical classification toward environmental risk modelling. Terrain is important because elevation and slope provide spatially varying physical context that can be associated with landslide susceptibility. The terrain module therefore extracts elevation and slope at every positive/background sample.

3.2 DEM Source and Processing

Copernicus DEM GLO-30 was used as the elevation source. Thirty $1^\circ \times 1^\circ$ DEM tiles were required to cover the 15,770 sample locations. The tiles were downloaded locally because direct remote COG streaming through the project's GDAL/Rasterio setup was unreliable.

Elevation was sampled at each point. Slope was derived from the DEM using a Horn-style 2D spatial gradient and reported in degrees.

```
Main terrain outputs:
data/processed/landslide_environmental_features.csv
data/processed/landslide_environmental_features_clean.csv
```

```
Terrain columns:
Elevation_m
Slope_deg
```

3.3 Terrain Extraction Results

Measure	Elevation	Slope
Valid samples	14,270 / 15,770	14,270 / 15,770
Missing samples	1,500	1,500
Overall range	-1.7 to 2454.7 m	0 to 63.98°
Median	599.4 m	12.85°
Mean	602.8 m	14.17°
IQR	187.5–820.2 m	3.97–22.68°
Landslide median	678.6 m	20.91°
Landslide mean	741.9 m	20.49°
Background median	523.6 m	4.68°
Background mean	469.2 m	8.10°

3.4 Missing DEM Investigation

The initial extraction produced 1,500 missing elevation/slope records. Missing values were investigated rather than being filled with arbitrary terrain estimates. A repair procedure searched for nearby valid DEM pixels within a local radius. Only 19 records could be repaired, while 1,481 remained without a valid DEM pixel within the tested 100-pixel (~3 km) search radius.

One diagnostic point at approximately latitude 9.0, longitude 77.092389 was found just outside the lower edge of its DEM tile, demonstrating that some missing values were boundary-related. However, because most missing points had no valid DEM pixel within the tested local radius, the remaining values were not fabricated or extrapolated.

3.5 Clean Terrain Dataset

For downstream modelling, rows without valid elevation and slope were removed. This produced a clean terrain dataset of **14,270 samples**.

Class	Count
Landslide	6,991
Background	7,279
Total	14,270

4. Files and Reproducibility

The implementation is being maintained as a reproducible Python project rather than as a collection of manually edited datasets. Large source files such as the original PDF and DEM tiles are intended to remain outside GitHub, while scripts document how they can be regenerated.

Component	Relevant file(s)
PDF extraction	src/extract_dataset.py
Preprocessing	src/preprocess_data.py
Inspection	src/inspect_dataset.py
Baseline RF	src/train_baseline_model.py
Background sampling	src/generate_background_samples.py
DEM extraction	src/extract_dem_features.py
DEM diagnostics	src/diagnose_dem_missing.py
DEM repair	src/repair_missing_dem_features.py
Processed binary data	data/processed/landslide_binary_dataset.csv
Clean terrain data	data/processed/landslide_environmental_features_clean.csv
Terrain summary	outputs/environmental_features_summary.txt
Terrain charts	outputs/charts/elevation_distribution.png; slope_distribution.png; western_ghats_elevation_slope_n

5. What Phase A + B Achieved

At the end of Phase B, the project has moved beyond a simple historical Random Forest classifier. The system now has a spatially defined Western Ghats study region, a balanced positive/background modelling dataset, and point-level terrain features for 14,270 samples.

Milestone	Completed result
Historical inventory	34,207 extracted records
Western Ghats positive set	7,885
Background set	7,885
Binary dataset	15,770 samples
Valid terrain dataset	14,270 samples
Terrain variables	Elevation + slope
Baseline model	Random Forest, 69.60% accuracy / 0.33 macro F1

Milestone	Completed result
Reproducible code	Extraction, preprocessing, sampling, terrain extraction and diagnostics

6. Methodological Limitations to Carry Forward

- The first Random Forest predicts historical movement type, not future landslide probability.
- The background class represents sampled non-event locations and should not automatically be interpreted as confirmed true negatives.
- The current Western Ghats polygon is a project study boundary derived from administrative boundaries and the inventory, not an official Western Ghats boundary.
- DEM missing values were not artificially imputed when a reliable local value could not be obtained.
- Historical event years are available for only a subset of records, so a common reference year is being used for the next environmental layers rather than pretending every sample has an event-year observation.
- The final model must use spatially aware train/validation/test splitting to reduce geographic leakage, especially because multiple historical records can share coordinates.
- Final evaluation should include ROC-AUC, PR-AUC, F1, recall and calibration rather than relying on accuracy alone.

7. Next Planned Work

Phase C will add NDVI, land cover and soil properties. AppEEARS requests have already been prepared in 15 batches because AppEEARS limits point requests to 1,000 coordinates. The NDVI/land-cover requests cover 2022. Phase D will then add daily rainfall and derive rolling rainfall features plus temporal sequences for the planned temporal Transformer.

Next phase	Planned implementation
Phase C	2022 MODIS NDVI + QA filtering, MCD12Q1 land cover, SoilGrids soil features
Phase D	CHIRPS v3 daily rainfall; 1/3/7/15/30-day aggregates; max 7/30-day rainfall; 30-day sequences
Phase E	Finalize integrated spatial dataset and validation/split strategy
Phase F	CNN spatial encoder + temporal Transformer + tabular encoder + feature fusion
Phase G	Compare RF, MLP, CNN, temporal model and multimodal fusion
Phase H	Risk probability mapping, dynamic rainfall update and alert/dashboard layer

Appendix A — Suggested Phase A/B Presentation Summary

One-minute explanation: We first extracted and cleaned a 34,207-record historical landslide inventory. We then restricted the study to a Western Ghats-focused region covering Maharashtra, Goa, Karnataka, Kerala and Tamil Nadu. Because historical landslide records contain only positive events, we generated 7,885 spatially separated background samples to construct a 15,770-sample binary dataset. A Random Forest was trained as a baseline on historical location/material information, achieving 69.60% accuracy but only 0.33 macro F1, showing the limitations of the imbalanced historical classification task. We then introduced terrain information using Copernicus DEM GLO-30, extracting elevation and slope. After excluding 1,500 samples with unresolved DEM values, the clean terrain dataset contains 14,270 samples. This terrain dataset is now the foundation for adding satellite vegetation, land cover, soil and rainfall features before developing the multimodal deep-learning model.

Appendix B — Important Terminology

Term	Use in Slidronix
Positive	Known historical landslide location from the inventory
Background	Sampled location inside the study region, spatially separated from positives
Binary target	Landslide = 1, Background = 0
Reference-year feature	Environmental feature calculated for a common year (currently 2022)
Terrain feature	Elevation or slope extracted from DEM
Early warning	Future system capability; should not be claimed as fully implemented until dynamic rainfall and validated temporal
Risk probability	Planned model output in the final multimodal system